

AMAN SINGANAMALA

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Full Stack Software Developer with experience building scalable web applications using React.js, Spring Boot, Java, JavaScript, TypeScript, and REST APIs in enterprise environments, with growing hands-on expertise in AI-driven development using LLMs, LangChain, and agentic application tooling. Proven expertise in authentication systems, role-based access control (RBAC), microservices architecture, and cloud-native deployments on Microsoft Azure (AKS, Azure Active Directory, Blob Storage). Strong background in designing secure, high-performance applications, implementing automated testing, and working in Agile/Scrum teams to deliver reliable, production-ready software solutions.

Technical Skills

Languages: Java, JavaScript (ES6+), TypeScript, Python, SQL

AI & LLM Tooling: LangChain, LLM Integration, Agentic Application Development, Model Context Protocol (MCP), GitHub Copilot, ChatGPT, Codex

Web Technologies: React.js, Spring Boot, Node.js, Microservices, REST APIs, HTML, CSS

Cloud & DevOps: Microsoft Azure (AKS, Azure Active Directory, Blob Storage), Docker, Kubernetes, Ansible

Databases & Tools: PostgreSQL, MongoDB, Git, GitLab, Postman, Linux, pgAdmin

Certifications: Microsoft Azure Fundamentals (AZ-900), UBS Certified Engineer

Professional Experience

UBS

August 2024 – Present

Software Developer (Full Stack Web Developer)

Pune, India

- Engineered a secure, scalable OTP-based authentication system using React.js and Spring Boot, implementing end-to-end OTP generation and validation with global state management to ensure consistent authentication behavior across multi-tab sessions, supporting 1,000+ daily authentication requests.
- Built modular, type-safe grid components and a dynamic records management module with advanced filtering and editing, improving UI responsiveness for real-time customer verification workflows.
- Implemented React UI test cases using Vitest, achieving 94% line code coverage and reducing UI regression issues.
- Architected an end-to-end Mutual Fund investment module by developing intuitive React dashboards for portfolio tracking and scalable Spring Boot APIs for secure transaction processing, supporting multiple concurrent user workflows.
- Engineered a robust Maker-Checker approval system for sensitive bulk uploads (CSV, XLSX), enforcing strict validation layers prior to database persistence and reducing data inconsistencies by 99%.
- Implemented enterprise-grade Role-Based Access Control (RBAC) using Azure Active Directory and Microsoft Graph API, securing API endpoints and enforcing authorized access for 5+ distinct user roles across internal applications.
- Developed an internal onboarding tool that centralized role-based access and permissions, reducing manual access coordination efforts by 50% and accelerating new-hire onboarding.
- Mentored junior engineers through task assignments and code reviews within an Agile/Scrum workflow, raising code quality standards and sprint execution efficiency across the team.

UBS

January 2024 – July 2024

Software Developer Intern (Full Stack Web Developer)

Pune, India

- Developed a full-stack application to manage client-specific pricing rates, supporting validation workflows during trade bookings and reducing manual verification efforts by 95%, including an early maker-checker validation layer later matured into the full-time role's approval system.
- Designed robust REST APIs and integrated frontend services using Axios, implementing secure handling of headers, authentication tokens, and error states to ensure stable client-server communication.
- Set up PostgreSQL databases for web applications, enabling reliable data storage and retrieval using Azure Cloud, Ansible, and pgAdmin.
- Orchestrated the containerization and deployment of web applications using Docker and Kubernetes on Azure Kubernetes Service (AKS); leveraged Ansible for configuration management, reducing environment setup time by 30%.

Projects

DSA Lab | *Next.js, React, TypeScript, Tailwind CSS, shadcn/ui, Supabase, PostgreSQL, Google OAuth, Vercel, NVIDIA NIM LLM API* **2026**

- Built DSA Lab (live demo), an interactive algorithm and data structure visualizer that helps students understand core CS concepts through step-by-step visual execution.
- Built DSA Lab (live demo), an interactive Next.js platform that visualizes algorithm execution step by step for arrays, hash maps, graphs, recursion, and sliding-window problems.
- Implemented Google OAuth and Supabase-backed user accounts, enabling learners to save visualizations by topic and revisit their practice history.
- Integrated NVIDIA NIM-powered translation and an in-app DSA tutor that provides grounded, step-by-step explanations from the current execution trace.

Education

Vellore Institute of Technology

August 2020 – June 2024

Bachelor of Technology in Computer Science, CGPA: 8.86 / 10.0

Vellore, India

Narayana Junior College

May 2018 – May 2020

High School, Percentage: 98.5%

Hyderabad, India

Awards & Leadership

- Recipient of Engineering Excellence Award at UBS for outstanding performance as a Graduate Trainee.
- Editorial Head, IEEE – Microwave Theory and Technique Society (MTTS), VIT.
- Content Writer, Centre for Social Entrepreneurship and Development (CSED), VIT.